Tit4Tat Trading Bot

An explainable autonomous trader for the MANTHAN 2026 challenge



Team Tit4Tat · portal ID MN2602

Grow ₹10 lakh in a live market, after tax

A live market stream

Prices and news arrive over Kafka; a simulated trading day passes every few seconds, with nobody at the keyboard.

Real costs and Indian tax

₹20 per order, ₹15.34 per sale, 15% tax on short-term gains, 10% on long-term gains above ₹1 lakh a year.

Strict platform rules

Heartbeats must be acknowledged in pre-flight, and a team that sends too many requests gets blocked.

₹10 L
starting cash

40
stocks in the final universe

~4 s
per simulated trading day

How the bot decides, every simulated day



1. Listen

Collects the day's prices and headlines from Kafka.

2. Measure

Trend, momentum, RSI, volume flow and a news score per stock.

3. Decide

Ranks all 40 stocks; the risk guard can cut size or stop buying.

4. Plan

Up to 10 orders, each checked against cash and platform rules.

5. Trade

Sends orders, books fills and checks the ledger against the server.

~400 ms
to decide each day, measured live

10
orders a day at most, highest priority first

Every order
logged with the reason it was placed

Own the strongest stocks, keep the tax low

Pick the best 10 of 40

Volume flow compares buying and selling volume over 50 days. Stocks where buyers dominate get more money.

Enter when buyers lead

A stock is bought only while its volume flow is positive. Any stock still waiting is bought by day 20.

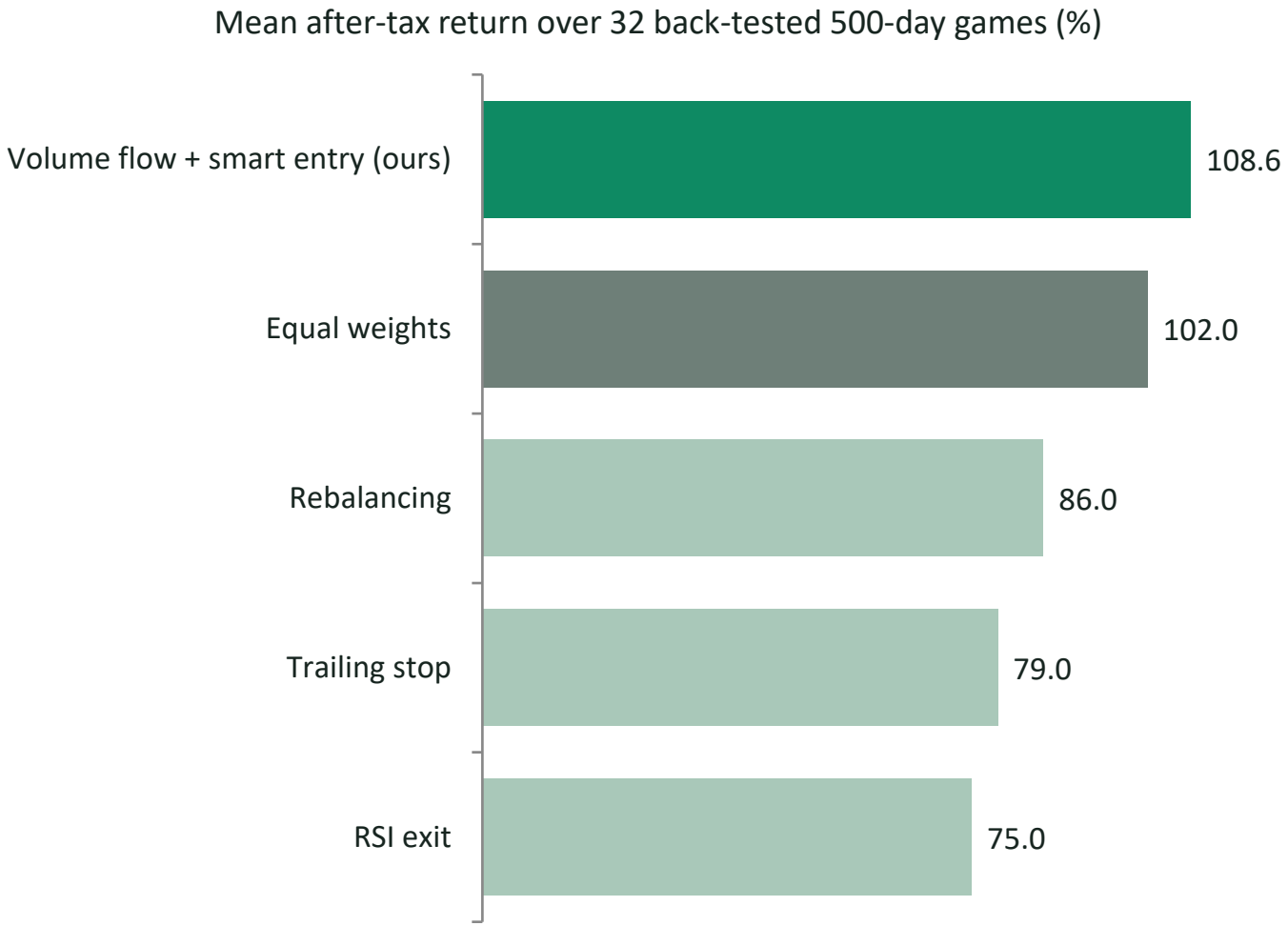
Hold and protect

The core is held through dips. No stock may exceed 30% of wealth, and an 8% drawdown halves new buying.

Keep the tax low

Uses the ₹1 lakh tax-free long-term gain every year, and books losses to cancel out taxable gains.

30+ experiments: we kept only what won



–23 to –34 pts

Selling winners early (RSI exits, trailing stops, rebalancing) lost to holding them.

56% vs 12.5%

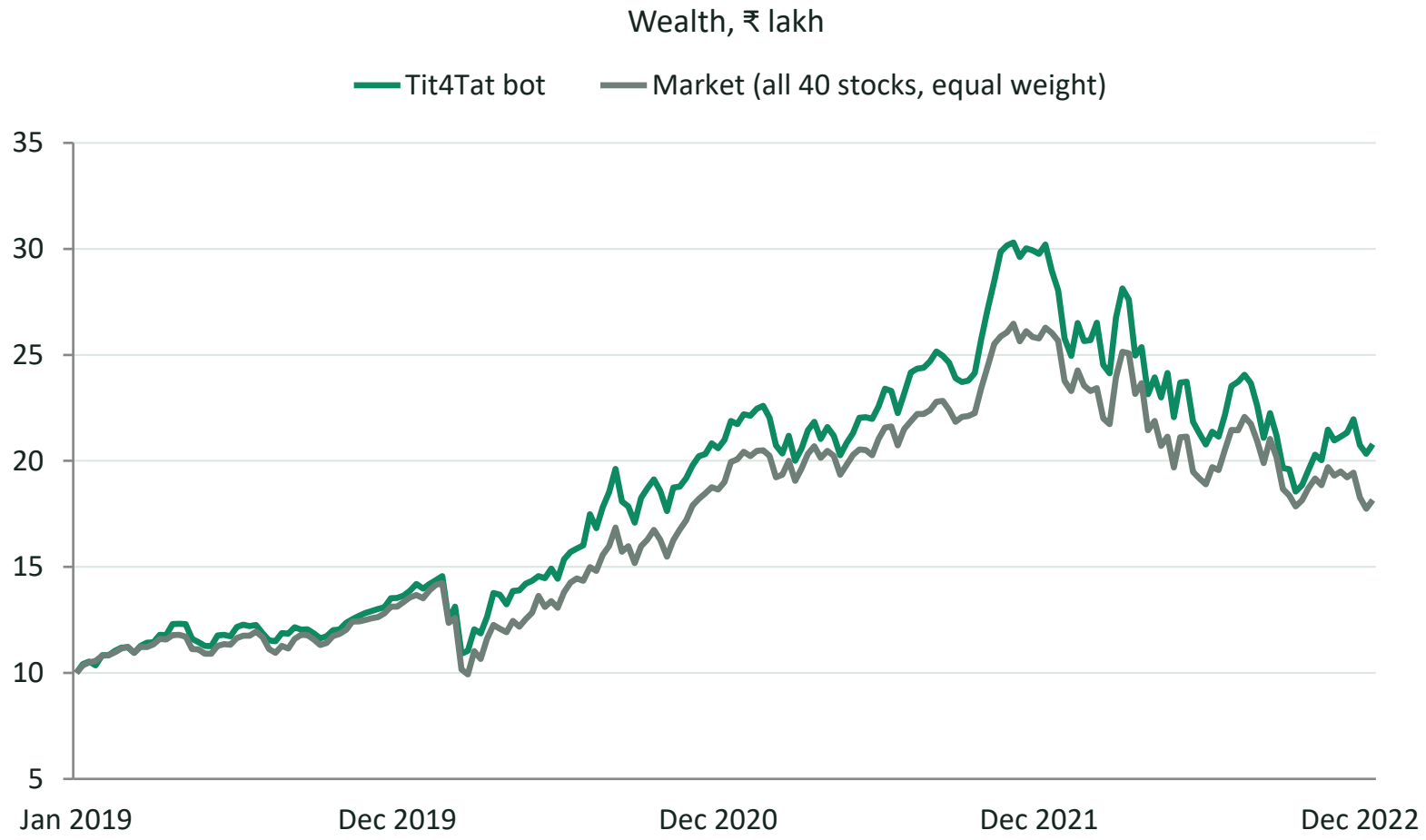
Games that beat buy-and-hold: volume weighting vs equal weights.

No next-day edge

Headlines moved prices on the same day, not the next, so news never triggers a trade.

Best setting of each idea shown. Every variant was replayed over every 500-day stretch of the history, with fees and tax.

Pre-flight test: +107.9% vs the market's +81.5%



₹20.79 L
final wealth after 1,008 trading days

₹31.1 L
peak, November 2021

41%
deepest fall, in the 2022 sell-off

140 / 2
orders filled / rejected

Official pre-flight run on the platform, 2 Jan 2019 – 30 Dec 2022, completed. Bot wealth is before an estimated ₹18.7k tax; market line is buy-and-hold of all 40 stocks.

Built to run unattended

96 automated tests

Strategy, tax, ledger, order checks and failure recovery.

Request pacer

At most 12 requests per 5 s, below the platform's limit of 15.

Kafka watchdog

Rejoins the stream if it gets no partitions for 20 s.

Heartbeats

Acknowledged in order and retried through server timeouts.

Server reconcile

The local ledger is checked against the server's cash and holdings.

Secrets kept out

The token lives only in .env; the submission ZIP is secret-scanned.

What we learned

1

Patience pays

Holding strong stocks beat every clever exit rule we tried.

2

Volume shows who is in control

Buying vs selling pressure picked better stocks than price alone.

3

Tax is part of the return

Tax-free gains and loss harvesting cost nothing and keep more profit.

Next: soften deep drawdowns like 2022 without selling the winners.

Thank you

Questions?